

Writing papers: references and data management

The TREESLab way for reference management & manuscripts' data curation

Alber Ipia¹ alber.ipia@inpe.br

Guilherme Mataveli ¹

Debora Dutra ¹

¹National Institute for Space Research



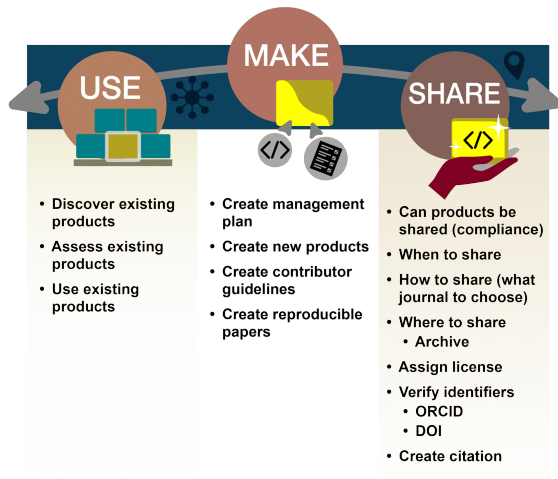
May 18, 2026

Agenda I

- 1 Introduction
- 2 Reference management
 - Reference management software
 - Zotero
 - Zotero and TREESLab
- 3 Research data management
 - Zenodo
 - Zenodo and TREESLab
- 4 Final remarks
- 5 References

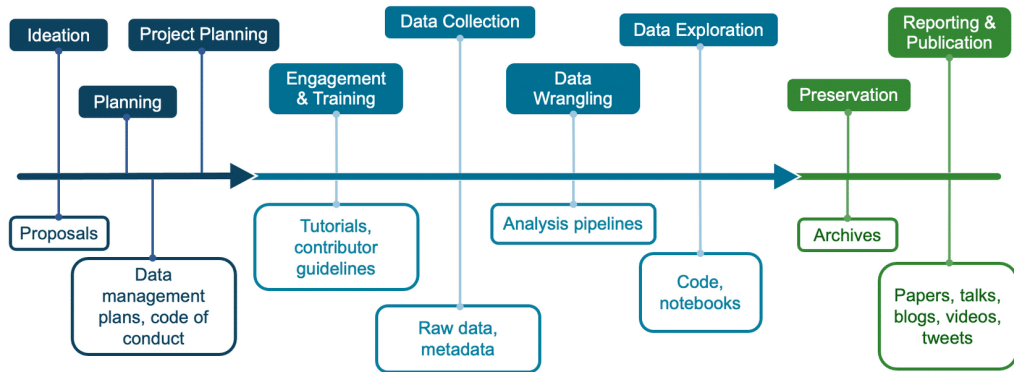
What do researchers do?

Use, make, and share open results



Source: OpenScience101.org

The research cycle



Source: OpenScience101.org

Introduction

- The scientific endeavour is time-consuming.
- Scientific research includes, among other tasks, keep track of the current scientific literature and use to foster new discoveries.
- But also, ensuring the scientific reproducibility of their experiments.
- Some work related to these task could done using specialized software and tools.

Digital Object Identifier DOI

- Unique.
- Identifier.
- Persistent.
- Resolvable.
- Metadata.
- FAIR:
 - ▶ Findable.
 - ▶ Accessible.
 - ▶ Interoperable.
 - ▶ Reusable.



Reference management.

Reference management

What do researches do with bibliographic references? [5]

- Import them from databases.
- Organize them.
- Gather additional metadata.
- Highlight and add notes to them.
- Sharing them.
- Convert them between formats or styles.
- Insert them to documents using word processing software.

Tools for reference management

- (Outdated) Comparison between CiteULike, RefWorks, Mendeley, and Zotero [5].
- For a more up-to-date comparison, also check [software comparison article in Wikipedia](#).

Zotero

- Free and open source reference management software (AGPL v3 license).
- Free cloud storage up to 300 MB.
- Released in 2006.
- Commonly used in medicine [2, 6].
- Integration with Word Processor Software (MS Word, LibreOffice Writer, OnlyOffice, and Google docs).
- Integration with Web Browsers (Firefox, Chrome, Safari, Edge).
- Download [here](#).



Import references to Zotero

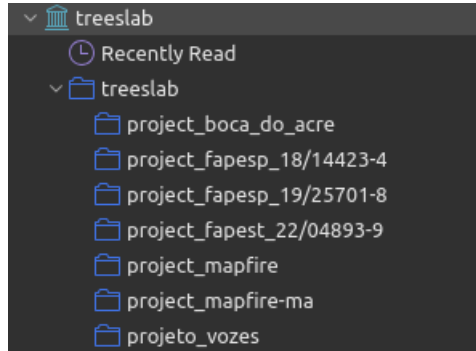
Import references from other software:

- Mendeley.
- EndNote.
- Citavi.
- MS Word bibliography XML.
- Plain text.
- Bib(La)Tex.
- JabRef.
- Check how [here](#).

The logo for Citavi, featuring the word "citavi" in a blue, lowercase, sans-serif font. A red comma-like shape is positioned below the "i" and above the "a".

Organize references into collections

- References can be organized into collections.
- Collections allow hierarchical organization.
- The same item (reference) can belong to multiple collections.
- Use drag and drop to add items to collections.
- The TREESLab library has a collection for each project.



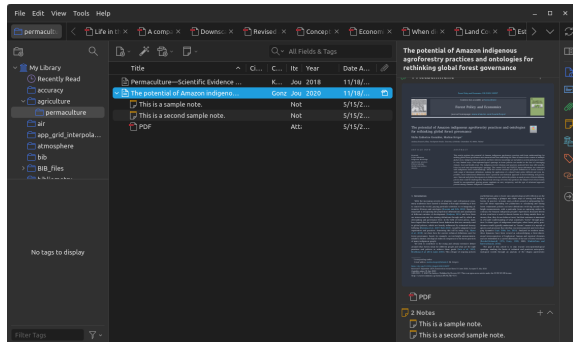
Gather additional metadata

- Zotero allows you to fill additional metadata for each item.
- You can even associate a PDF to an item (reference) in Zotero.

Amazon forest response to repeated droughts	
Info	
Citation Key	
Item Type	Journal Article
Title	Amazon forest response to repeated droughts
Author	Feldpausch, T.R.
Author	Phillips, O.L.
Author	Brienen, R.J.W.
Author	Gloor, E.
Author	Lloyd, J.
	51 more...
Publication	Global Biogeochemical Cycles
Publisher	
Place	
Date	07/2016
Volume	30
Issue	7
Section	
Part Number	
Part Title	
Pages	964-982
Series	
Series Title	
Series Text	
Journal Abbr	Global Biogeochemical Cycles
DOI	10.1002/2015GB005133
URL	https://agupubs.onlinelibrary.wiley.com/doi/10.1002/2015GB005133
Accessed	11/28/2023, 12:54:14 PM
PMID	
PMCID	
ISSN	0886-6236, 1944-9224
Archive	
Loc. in Archive	
Short Title	
Language	en
Library Catalog	DOI.org (Crossref)
Call Number	
License	
Extra	
Date Added	12/11/2023, 8:05:08 PM
Added By	Alber Sánchez

Highlight and add notes to references

- Zotero allows you to highlight and add notes to a PDF associated to an item (reference).
- Notes help [here](#).



Share references

- Zotero allows you to create groups and use them to share references with colleagues.
- The TREESLab' publication list is a collection associated with the Zotero group called *treeslab*.

treeslab

[Group Library](#)

[Manage Profile](#) · [Manage Members](#) · [Manage Library](#)

Members 22

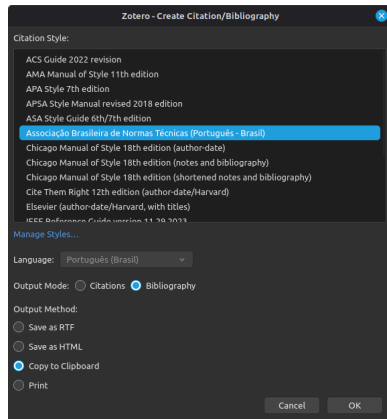
Description TREES laboratory is a research group co-led by Dr Luiz Aragão and Dr Liana Anderson. The group was created in 2009 when Dr Aragão moved from the Environmental Change Institute, University of Oxford to the School of Life and Environmental Sciences, University of Exeter in UK. Dr Aragão is now based at the National Institute for Space Research in Brazil, the current headquarters of TREES. Dr Liana Anderson is currently based at The National Center for Monitoring and Early Warning of Natural Disasters (CEMADEN), Brazil. The group's expertise is on assessing the risks and impacts of environmental changes on tropical ecosystems using remote sensing and field surveys. TREES members are trained in a multidisciplinary range of techniques and a multitude of instrumentation, technologies and approaches.

Group Type Public, Closed Membership

Group Library Anyone can view, only admins can edit

Convert references between formats or styles

- Zotero allows you to export items (references) or collections to several formats.
- It also allows to copy references and format them according to several reference styles.
- It is possible to add the ABTN style from [here](#).

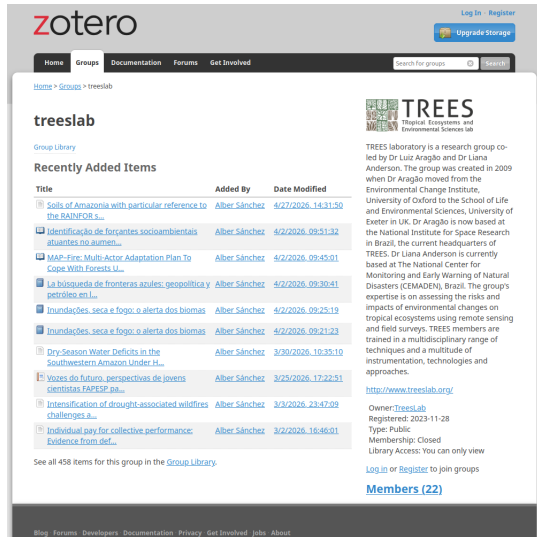


Insert references into text documents

- MS Word.
- Libre Office.
- Google Docs (using the Zotero connector for Web Browsers).
- LaTeX (export collection to a BIB file).

TREESLab group at Zotero

- TREESLab has a Zotero group (click [here](#)).
- This group hosts TREESLab's publication list.
- The group owner is TREESLab's email account wertreeslab@gmail.com
- Here we keep track of the publications produced by TREESLab.
- TREESLab members don't have to be members to add their publications to the list (but it'd be nice, though).



The screenshot shows the Zotero web interface for the 'treeslab' group. At the top, the Zotero logo is on the left, and 'Log In' and 'Register' links are on the right. Below the logo is a navigation bar with 'Home', 'Groups', 'Documentation', 'Forums', and 'Get Involved'. A search bar is on the right of the navigation bar. The main content area is titled 'treeslab' and includes a 'Group Library' link. Below this is a 'Recently Added Items' section with a table of publications. To the right of the table is a profile for 'TREES' (Tropical Ecosystems and Environmental Sciences lab), including a description of the laboratory and its location. At the bottom right, there is information about the group owner, registration date, type, membership status, and library access. A footer at the very bottom contains links for 'Blog', 'Forums', 'Developers', 'Documentation', 'Privacy', 'Get Involved', 'Jobs', and 'About'.

zotero Log In Register Upgrade Storage

Home Groups Documentation Forums Get Involved Search for groups Search

Home > Groups > treeslab

treeslab
Group Library

Recently Added Items

Title	Added By	Date Modified
Soils of Amazonia with particular reference to the RAINFOR s...	Alber Sánchez	4/27/2026, 14:31:50
Identificação de forçantes socioambientais atuantes no aumen...	Alber Sánchez	4/27/2026, 09:51:32
MAP-Fire: Multi-Actor Adaptation Plan To Cope With Forests U...	Alber Sánchez	4/27/2026, 09:45:01
La búsqueda de fronteras azules: geopolítica y petróleo en L...	Alber Sánchez	4/27/2026, 09:30:41
Inundações, seca e fogo: o alerta dos biomas	Alber Sánchez	4/27/2026, 09:25:19
Inundações, seca e fogo: o alerta dos biomas	Alber Sánchez	4/27/2026, 09:21:23
Dry-Season Water Deficits in the Southwestern Amazon Under H...	Alber Sánchez	3/30/2026, 10:35:10
Vozes do futuro, perspectivas de jovens cientistas FAPESP pa...	Alber Sánchez	3/25/2026, 17:22:51
Intensification of drought-associated wildfires challenges a...	Alber Sánchez	3/3/2026, 23:47:09
Individual pay for collective performance: Evidence from def...	Alber Sánchez	3/2/2026, 16:46:01

See all 458 items for this group in the [Group Library](#).

TREES
Tropical Ecosystems and Environmental Sciences lab

TREES laboratory is a research group coled by Dr Luiz Aragão and Dr Liana Anderson. The group was created in 2009 when Dr Aragão moved from the Environmental Change Institute, University of Oxford to the School of Life and Environmental Sciences, University of Exeter in UK. Dr Aragão is now based at the National Institute for Space Research in Brazil, the current headquarters of TREES. Dr Liana Anderson is currently based at The National Center for Monitoring and Early Warning of Natural Disasters (CEMADEN), Brazil. The group's expertise is on assessing the risks and impacts of environmental changes on tropical ecosystems using remote sensing and field surveys. TREES members are trained in a multidisciplinary range of techniques and a multitude of instrumentation, technologies and approaches.

<http://www.treeslab.org/>

Owner: [Treeslab](#)
Registered: 2023-11-28
Type: Public
Membership: Closed
Library Access: You can only view

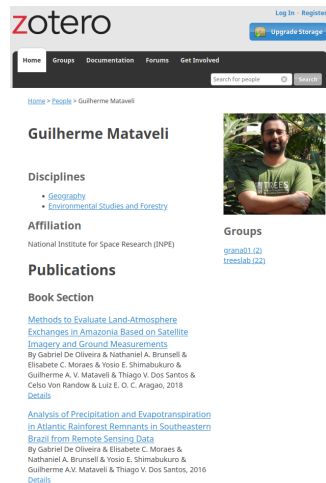
[Log in](#) or [Register](#) to join groups

[Members \(22\)](#)

Blog Forums Developers Documentation Privacy Get Involved Jobs About

Researcher's profile at zotero.org

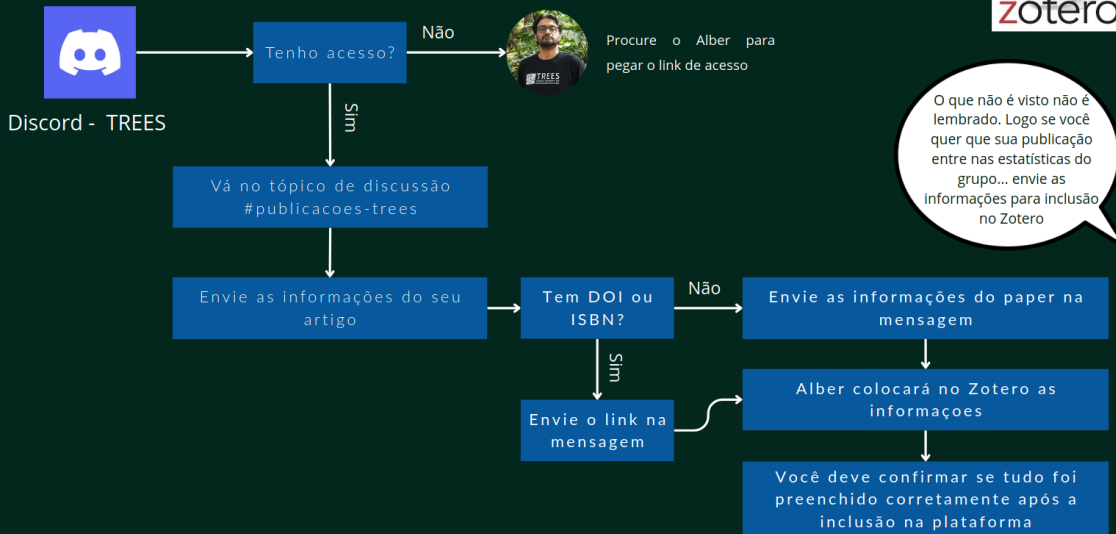
- Researchers can build a profile and zotero.org will host it.
- This profile would display the items in the *My Publications* collection (check [here](#)).



The screenshot shows the profile of Guilherme Mataveli on the Zotero.org website. The header includes the Zotero logo, links for 'Log In' and 'Register', and a button for 'Upgrade Storage'. The navigation bar contains links for 'Home', 'Groups', 'Documentation', 'Forums', and 'Get Involved', along with a search bar. The profile page shows the name 'Guilherme Mataveli' and a profile picture. Under 'Disciplines', it lists 'Geography' and 'Environmental Studies and Forestry'. The 'Affiliation' is 'National Institute for Space Research (INPE)'. The 'Groups' section lists 'gran01 (2)' and 'treelab (22)'. The 'Publications' section includes a 'Book Section' titled 'Methods to Evaluate Land-Atmosphere Exchanges in Amazonia Based on Satellite Imagery and Ground Measurements' by Gabriel De Oliveira & Nathaniel A. Brunsell & Elisabete C. Moraes & Yosio E. Shimabukuro & Guilherme A. V. Mataveli & Thiago V. Dos Santos & Celso Von Randow & Luiz E. O. C. Aragao, 2018. Another publication listed is 'Analysis of Precipitation and Evapotranspiration in Atlantic Rainforest Remnants in Southeastern Brazil from Remote Sensing Data' by Gabriel De Oliveira & Elisabete C. Moraes & Nathaniel A. Brunsell & Yosio E. Shimabukuro & Guilherme A.V. Mataveli & Thiago V. Dos Santos, 2016.

A researcher's profile. Source: zotero.org.

Publiquei meu artigo e quero colocar na biblioteca do TREES, o que fazer?



Research data management.

What do researchers do with the data associated to their publications?

Zenodo

- Open repository launched in 2013 and run by [CERN](#).
- Let researchers upload files up to 50 GB.
- It assigns a DOI to uploaded datasets.
- It supports GitHub repositories.
- It was named after Zenodotus, the first librarian of the Library of Alexandria [1].



Why use Zenodo?

Data could be lost due to many reasons:



(a) Carelessness.



(b) Handling.



(c) Catastrophe.

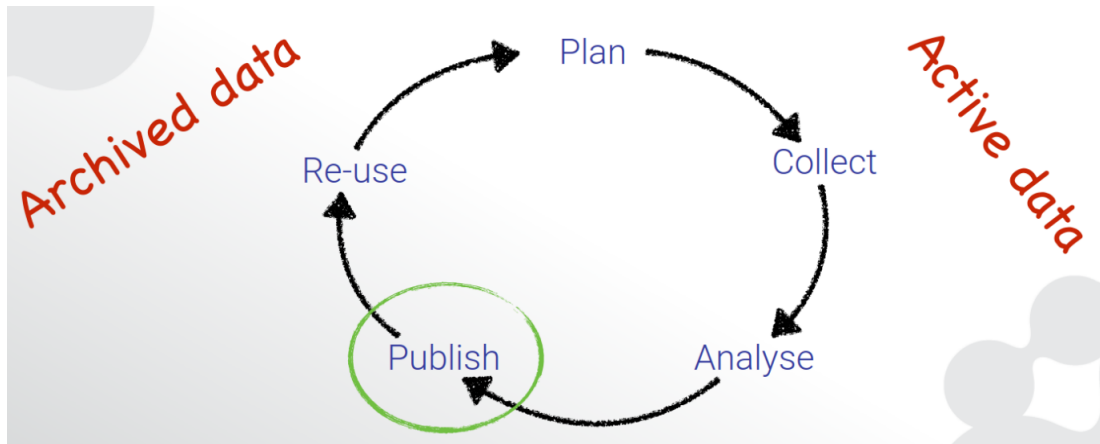
Examples of data loss [3].

Why use Zenodo?

- **Safe:** Data stored at CERN's data centre.
- **Trusted:** Operated by CERN.
- **No waiting time:** Available as soon as published.
- **Open or closed:** Restricted access mode.
- **Usage statistics.**
- **Versioning:** Update data and version numbers.
- **GitHub integration:** Preserve code.
- **Citeable:** Every upload has a DOI.
- **Data principles:** Findable, Accessible, Interoperable, Reusable.



Research data life cycle



Research data life cycle according [3].

How do you do it?

Sharing your data and software on Zenodo in 3 steps [3]:

- 1 Upload.
- 2 Describe.
- 3 Publish.

By the way, the documents cited here, are themselves hosted at Zenodo!



Source: Poster presented in Dublin [4].

Upload

Before you upload, think about:

- Data management plan.
- File formats (open versus proprietary).
- Standards.
- Re-usability.
- Licenses.
- Privacy.

Zenodo allows:

- Any format.
- Any licence.
- Maximum of 50GB* and 100 files per dataset.
- Drag and drop upload.

Describe

Basic information:

- Community (e.g. *treeslab*).
- Authors or creators.
- Type: Dataset, image, model, poster, presentation, publication (book, chapter, conference paper or proceeding, report), software (notebook), video, audio.
- Title.
- Licenses.

Recommended information:

- Contributors.
- Keywords and subjects.
- Dates.
- Version.
- Publisher.

Other information:

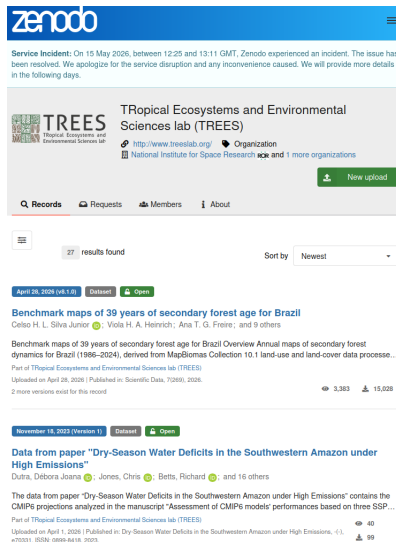
- Funding, alternate identifiers, related works, references, software, publishing information, conference, domain specific fields, visibility (public or restricted-embargo).

Publish

- Files are not editable.
- Metadata is editable.

TREESLab community at Zenodo

- TREESLab has a Zenodo community (click [here](#)).
- TREESLab members **must** have an Zenodo account to be able to submit datasets to this community.
- Submissions are reviewed for quality and compliance to TREESLab standards.
- There is a check list for submissions to TREESLab community (check it [here](#)).



The screenshot shows the Zenodo website interface. At the top, a blue header contains the Zenodo logo and a menu icon. Below the header, a white box contains a "Service Incident" notice dated May 15, 2026. The main content area features the "TREESLab" community profile, including its logo, name, website URL, and a list of member organizations. A "New upload" button is visible. Below the profile, a search bar shows "27 results found" and a "Sort by" dropdown set to "Newest". Two dataset listings are shown. The first dataset, titled "Benchmark maps of 39 years of secondary forest age for Brazil", was uploaded on April 28, 2026, and has 3,383 views and 15,028 downloads. The second dataset, titled "Data from paper 'Dry-Season Water Deficits in the Southwestern Amazon under High Emissions'", was uploaded on April 1, 2026, and has 40 views and 99 downloads.

zenodo

Service Incident: On 15 May 2026, between 12:25 and 13:11 GMT, Zenodo experienced an incident. The issue has been resolved. We apologize for the service disruption and any inconvenience caused. We will provide more details in the following days.

TREESLab
Tropical Ecosystems and Environmental Sciences lab (TREESLab)

<http://www.treeslab.org/> Organization
National Institute for Space Research and 1 more organizations

New upload

Records Requests Members About

27 results found Sort by Newest

April 28, 2026 (v8.1.0) Dataset Open

Benchmark maps of 39 years of secondary forest age for Brazil
Celso H. L. Silva Junior; Viola H. A. Heinrich; Ana T. G. Freire; and 9 others

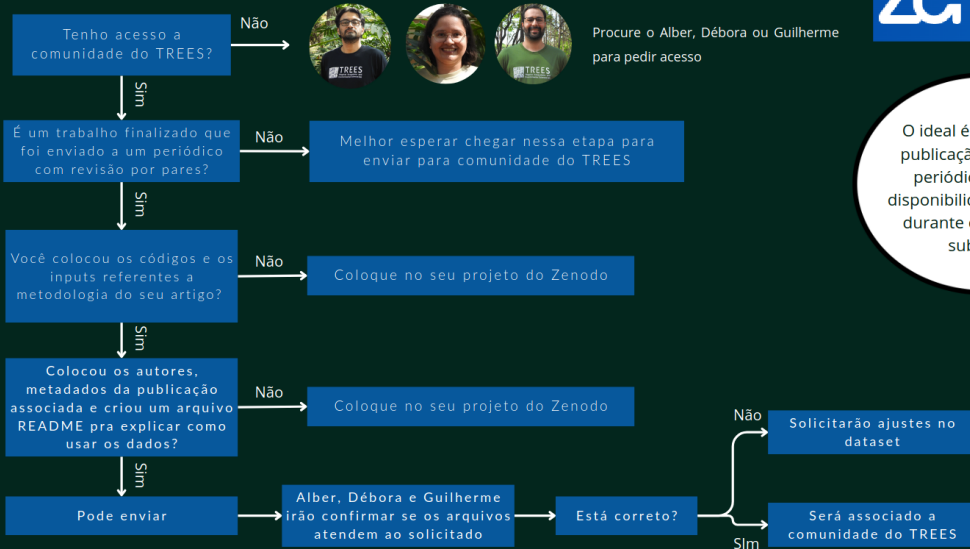
Benchmark maps of 39 years of secondary forest age for Brazil Overview Annual maps of secondary forest dynamics for Brazil (1986–2024), derived from MapBiomas Collection 10.1 land-use and land-cover data processed...
Part of Tropical Ecosystems and Environmental Sciences lab (TREESLab)
Uploaded on April 28, 2026 | Published in: Scientific Data, 7(269), 2026.
2 more versions exist for this record 3,383 15,028

November 18, 2023 (Version 1) Dataset Open

Data from paper "Dry-Season Water Deficits in the Southwestern Amazon under High Emissions"
Dutra, Débora Joana; Jones, Chris; Betts, Richard; and 16 others

The data from paper "Dry-Season Water Deficits in the Southwestern Amazon under High Emissions" contains the CMIP6 projections analyzed in the manuscript "Assessment of CMIP6 models' performances based on three SSP ..."
Part of Tropical Ecosystems and Environmental Sciences lab (TREESLab)
Uploaded on April 1, 2026 | Published in: Dry-Season Water Deficits in the Southwestern Amazon under High Emissions, <1>. e70331, ISSN: 0859-6416, 2023. 40 99

Quero associar um dataset de uma publicação no zenodo do TREES



O ideal é enviar após a publicação, mas alguns periódicos pedem a disponibilidade do dataset durante o processo de submissão

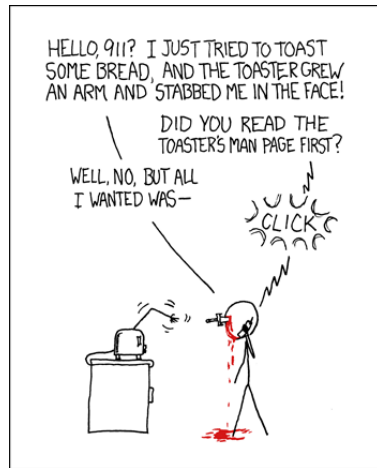
Final remarks.

Final remarks

- Please wonder if it is worth it to learn these tools, especially if you are going to be doing science for the rest of your life.
- Check [here](#) the *Tutorial para gerenciamento de referências bibliográficas utilizando o software Zotero* by the Ibict.
- There is a check list for submissions to TREESLab's Zenodo community (check it [here](#)).
- Also, check TREESLab' Discord chanel:
 - ▶ [ajuda-zotero](#).
 - ▶ [ajuda-zenodo](#).

RTFM

- Users don't acknowledge RTFM as their first option to solve software problems.
- This is so common, that software developers created RTFM as a gentle reminder to users of their options when dealing with software they don't know (check Wikipedia's article [about it](#)).



Source: RTFM by [xkcd](#).

References

- [1] European Organization For Nuclear Research and OpenAIRE. *Zenodo*. CERN, 2013. DOI: [10.25495/7GXX-RD71](https://doi.org/10.25495/7GXX-RD71).
- [2] K. K. Mueen Ahmed and Bandar E. Al Dhubaib. "Zotero: A Bibliographic Assistant to Researcher". In: *Journal of Pharmacology and Pharmacotherapeutics* 2.4 (Dec. 2011), pp. 304–305. DOI: [10.4103/0976-500X.85940](https://doi.org/10.4103/0976-500X.85940).
- [3] Lars Holm Nielsen. "Sharing Your Data and Software on Zenodo". In: (May 22, 2017). DOI: [10.5281/ZENODO.802100](https://doi.org/10.5281/ZENODO.802100).
- [4] Lars Holm Nielsen and Tim Smith. "Zenodo Overview". In: (Mar. 26, 2014). DOI: [10.5281/ZENODO.8428](https://doi.org/10.5281/ZENODO.8428).
- [5] Et Al. Ron Gilmour. *Reference Management Software: A Comparative Analysis of Four Products*. 2011. DOI: [10.5062/F4Z60KZF](https://doi.org/10.5062/F4Z60KZF). Pre-published.
- [6] Thomas E. Vanhecke. "Zotero". In: *Journal of the Medical Library Association : JMLA* 96.3 (July 2008), pp. 275–276. DOI: [10.3163/1536-5050.96.3.022](https://doi.org/10.3163/1536-5050.96.3.022).